Student Dropout Prediction: A Machine Learning Analysis

DSAA2011 (L01) Course Project — HKUST(GZ) 2026 Spring

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Abstract

We analyze the UCI Student Dropout and Academic Success dataset (4,424 samples, 36 features, 3 classes) using a complete machine learning pipeline encompassing data preprocessing, dimensionality reduction, clustering, supervised classification, and model evaluation. Our best-performing model achieves competitive F1-scores through hyperparameter tuning guided by validation curves. We find that academic performance features are the strongest predictors, and the "Enrolled" class is consistently the hardest to classify.

1 Introduction

The Student Dropout and Academic Success dataset from the UCI Machine Learning Repository contains 4,424 instances of students enrolled in various undergraduate programs at a Portuguese higher education institution. Each instance is described by 36 features covering demographics, socioeconomic background, and academic performance across the first and second semesters. The classification target has three categories: **Dropout** (32.1%), **Enrolled** (18.0%), and **Graduate** (49.9%).

Our objectives are: (1) explore data structure through visualization and clustering; (2) train and evaluate supervised classifiers; (3) improve models via validation-driven tuning; (4) investigate feature importance and ensemble methods.

Our best-performing model, Random Forest with default hyperparameters (`n_estimators=100`, `class_weight='balanced'`), achieves a macro F1-score of 0.7107 on the held-out test set. A key challenge is the "Enrolled" class, which represents only 18% of the data and is inherently ambiguous. By applying class-balanced weighting, we improve Enrolled class F1 from 0.00–0.52 (unweighted) to 0.47–0.52 (balanced) across all model types. Feature importance analysis confirms that academic performance metrics (approval rates, semester grades) are the dominant predictive signals.

2 Mandatory Tasks

2.1 Data Preprocessing

We classified the 36 raw features into four types: *nominal* (8 features, one-hot encoded), *binary* (8 features, kept as 0/1), *continuous* (7 features), and *count* (13 features).

Feature engineering produced 25 additional domain features across six groups: academic ratios, socioeconomic indicators, financial risk flags, demographic categories, macroeconomic indices, and application behavior features. Notably, our subsequent feature importance analysis confirmed that these engineered metrics provided the strongest predictive signals. The top five features overall were all engineered academic ratios: `overall_approval_rate` (importance=0.1002), `approval_rate_2nd` (0.0691), `total_approved` (0.0504), `approval_rate_1st` (0.0448), and `grade_per_evaluation_2nd` (0.0421). Out-of-fold target encoding was used for high-cardinality categorical features to prevent data leakage.

StandardScaler was applied to all 45 numerical features (20 original continuous / count features and the 25 newly engineered domain features) to ensure uniform scale without affecting binary indicators. Feature selection retained the top-50 features by tree-based importance (out of 61 candidates after target encoding).

Missing values: The dataset contains zero missing values across all 4,424 samples and 36 features, so no missing value handling methods were necessary.

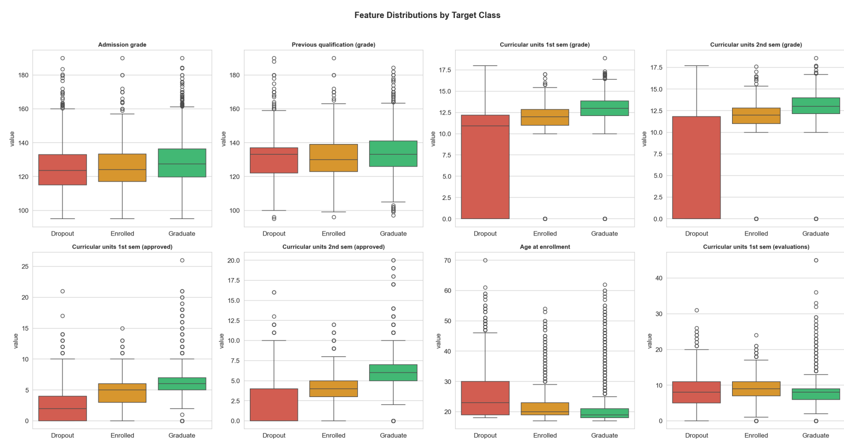

Figure 1: Feature distributions by target class.

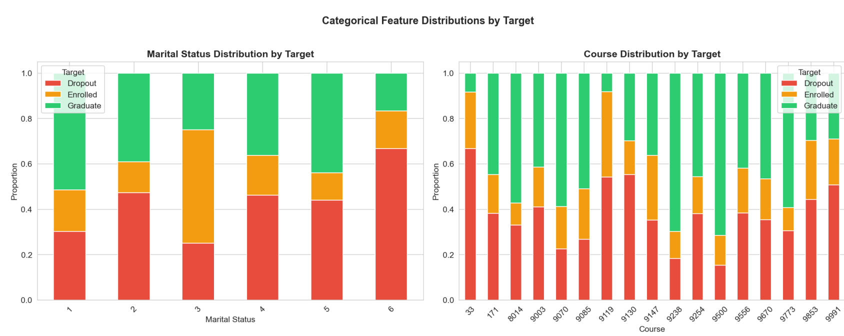

Figure 2: Categorical feature distributions (Marital Status and Course) stacked by target class.

Exploratory Data Analysis (EDA) reveals several key patterns:

- **Academic performance differences:** Graduate students have notably higher 1st/2nd semester grades than Dropout students, while the initial Admission grade shows less discrimination (Figure 1).
- **Age factor:** Dropout students tend to be older, showing a higher median age at enrollment.
- **Strongest predictive signals:** Curricular units approved and semester grade metrics show the strongest correlation with the target variable ($|r| > 0.3$) (Figure 3).
- **Demographics and program variations:** Different academic programs (Course) and demographic factors (e.g., Marital Status) exhibit vastly different dropout rates, as illustrated by the stacked distributions in Figure 2.

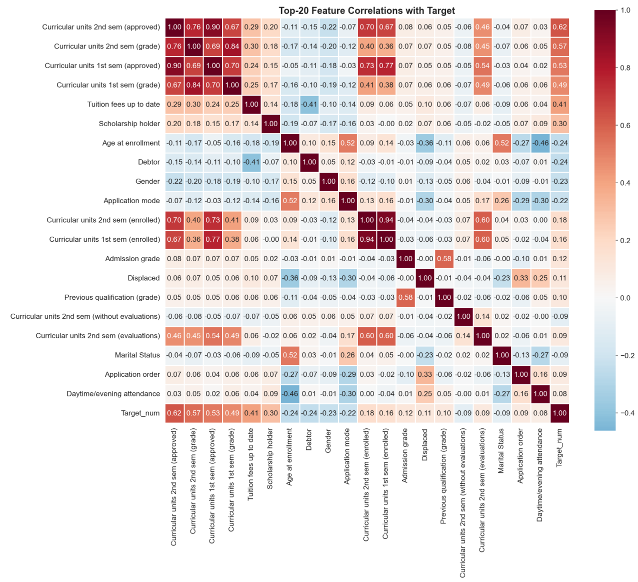

Figure 3: Top-20 feature correlations with the target variable.

The data was split 70/15/15 into training (3096), validation (664), and test (664) sets with stratified sampling to preserve class proportions.

2.2 Data Visualization (t-SNE)

To preserve the integrity of our unsupervised analysis and prevent data leakage, we applied t-SNE to the baseline preprocessed dataset (256 features, all 4,424 samples) rather than the engineered dataset, which incorporates target-encoded features that embed supervised label information.

We applied t-SNE to project the high-dimensional feature space into lower dimensions. A systematic pattern analysis across four perplexity values (Figure 4) highlights the algorithm's sensitivity to local versus global structure. At a low perplexity ($p = 5$), the projection is severely fragmented, obscuring any meaningful global patterns. Conversely, a high perplexity ($p = 100$) over-smooths the manifold, causing a loss of fine-grained local detail while significantly increasing computational overhead.

Figure 4: t-SNE 2D projections at perplexity = 5, 30, 50, 100.

Perplexity=30 achieves the optimal balance. The 2D projection (Figure 5, left) reveals that Graduate students form a relatively compact region, while Dropout students cluster separately. Enrolled students are scattered between both groups, consistent with their transitional academic status. Furthermore, extending the projection from 2D to 3D (Figure 5, right) provides additional spatial volume, further reducing the apparent overlap between the distinct Graduate and Dropout clusters.

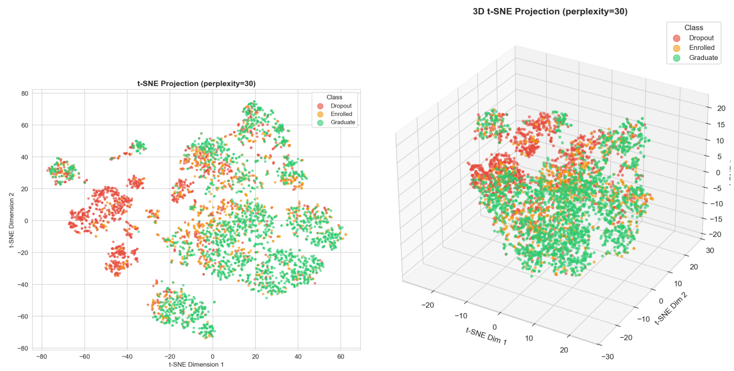

Figure 5: Left: t-SNE 2D projection (perplexity=30) colored by class. Right: t-SNE 3D projection (perplexity=30) colored by class.

2.3 Clustering Analysis

Consistent with the t-SNE analysis, clustering was also performed on the full baseline preprocessed dataset (256 features, 4,424 samples). Using the target-engineered feature set would introduce label information into the feature space, artificially inflating cluster purity metrics and violating the unsupervised setting.

Three algorithms were compared: K-Means, Agglomerative Hierarchical Clustering (ward, complete, average, single linkage), and DBSCAN.

K-Means Clustering: We evaluated K=2 through K=10 using Silhouette, Calinski-Harabasz (CH), and Davies-Bouldin (DB) scores, alongside the Elbow method (Figure 6). K=3 was selected as it aligns with the target class count and shows a distinct elbow in inertia reduction.

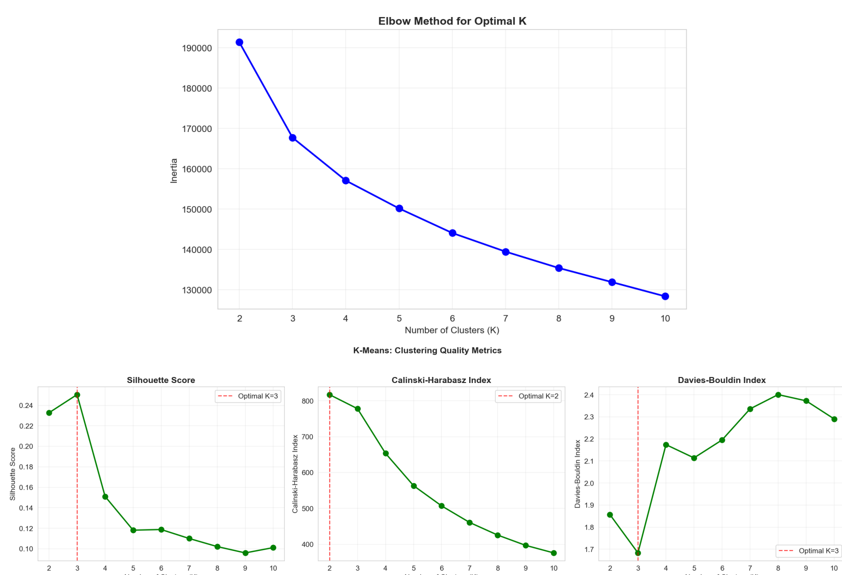

Figure 6: Top: Elbow method for K selection. Bottom: Silhouette, CH, and DB evaluation metrics vs. K.

K-Means (K=3) achieved the highest practical utility with an Adjusted Rand Index (ARI) of 0.177, producing reasonably balanced cluster sizes that correspond meaningfully to the three student categories. Cross-tabulation between clusters and true labels revealed significant structural insights: Cluster 1 is predominantly composed of Dropout students (82.7%), whereas Clusters 0 and 2 both capture predominantly Graduate students ($\sim$60%). This indicates that while dropouts form a relatively tight and distinct cluster, enrolled and graduate students share highly overlapping feature spaces. Figure 7 illustrates the cluster assignments mapped onto the t-SNE projection alongside the cluster size distribution.

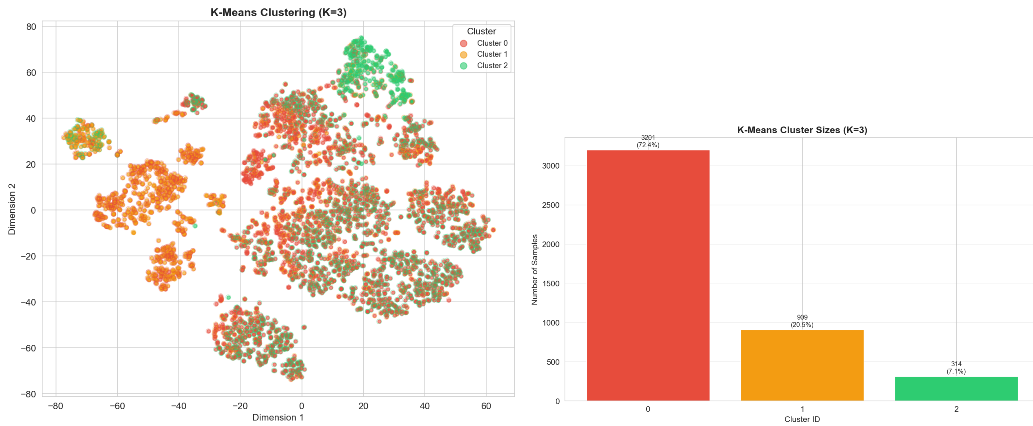

Figure 7: Left: K-Means (K=3) clusters on t-SNE projection. Right: K-Means cluster size distribution.

Hierarchical Clustering: The dendrogram for Ward linkage (Figure **??**) shows a relatively flat hierarchical structure, supporting K-Means' spherical cluster assumption. Among the four linkage methods, Ward achieved the best ARI (0.084). A visual and metric comparison (Figure **??**) revealed that single and average linkages were not just imbalanced, but completely uninformative. They produced one massive cluster alongside isolated outliers, yielding ARIs near zero (single: -0.0005, average: -0.0002). Conversely, Ward and complete linkages generated more balanced and meaningful partitions.

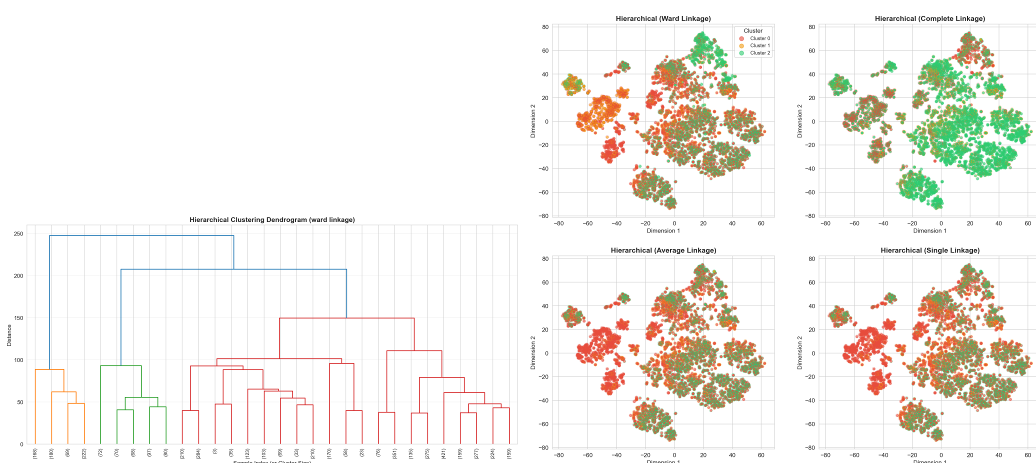

Figure 8: Left: Hierarchical clustering dendrogram (Ward linkage). Right: Comparison of clusters generated by Ward, Complete, Average, and Single linkages.

DBSCAN Analysis: To systematically configure DBSCAN, we plotted the k-distance graph (Figure 9) to identify the optimal ϵ parameter at the point of maximum curvature. Despite conducting a comprehensive grid search across multiple `eps` and `min_samples` combinations, DBSCAN failed to discover meaningful structures. The high-dimensional nature of the dataset caused severe distance concentration—a manifestation of the curse of dimensionality. This resulted in an extreme noise ratio where up to 99.8% of data points were classified as noise (ARI=0.000), confirming that density-based methods are unsuitable for this specific feature space without more aggressive non-linear dimensionality reduction.

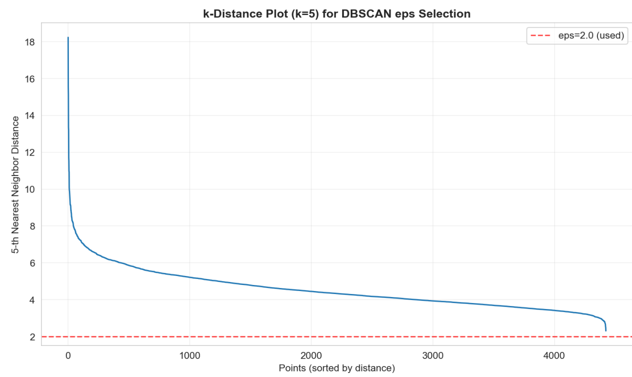

Figure 9: k-distance plot used for DBSCAN ϵ estimation.

Overall Algorithm Comparison: To synthesize our findings, centroid-based and variance-minimizing approaches proved significantly more robust for this tabular dataset than density-based methods. As summarized in Figure 10, K-Means (K=3) achieved the highest overall utility (ARI = 0.177) by successfully isolating the majority of Dropout students. Hierarchical clustering with Ward linkage provided a comparable, albeit slightly less distinct, partition (ARI = 0.084). In stark contrast, DBSCAN's failure (ARI = 0.000) highlights its vulnerability to the curse of dimensionality in our feature space, where distance concentration effectively obscures any natural density peaks. Ultimately, while K-Means provides the best macro-level segmentation, the inherent feature overlap between Enrolled and Graduate students remains the primary bottleneck across all unsupervised learning approaches.

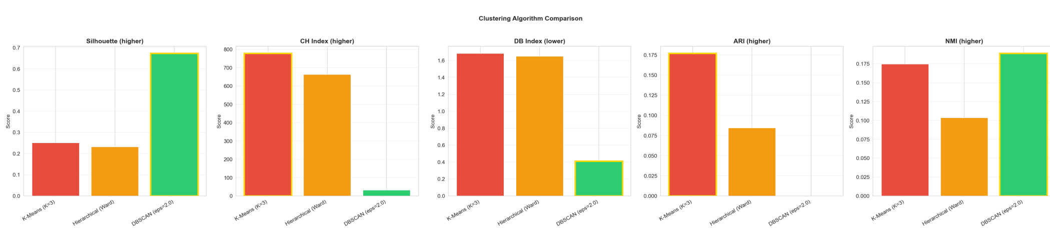

Figure 10: Clustering algorithm comparison across metrics.

2.4 Prediction: Training and Testing

Data preparation. We used the engineered feature dataset (50 features after OOF target encoding and top-50 selection), split into train (3,096), validation (664), and test (664) sets. The class distribution in the training set is: Dropout 32.1%, Enrolled 17.9%, Graduate 49.9%.

We trained three classifiers using default configurations from `prediction.py`: Decision Tree (`max_depth=10, class_weight='balanced'`), Logistic Regression (`solver='saga', class_weight='balanced'`), and SVM with RBF kernel (`class_weight='balanced'`). All models were evaluated on training, test, and entire sets.

Three-set evaluation. Table 1 reports accuracy and macro F1 across all evaluation sets. DT shows the largest train-test gap, indicating significant overfitting from its deep tree structure. LR provides stable generalization. SVM achieves the closest train-test alignment but at a lower overall performance level.

Table 1: Baseline model performance across evaluation sets.

Model	Train Acc.	Test Acc.	Entire Acc.	Test F1 (macro)
Decision Tree	0.9047	0.7154	0.8472	0.6795
Logistic Reg.	0.7594	0.7364	0.7545	0.7027
SVM (RBF)	0.6854	0.6867	0.6829	0.6449

Figure 11: Model performance across train, validation, test, and entire sets.

The per-class analysis shows that all models achieve the highest recall for Graduate and the lowest for Enrolled. This is consistent with the class imbalance and the transitional nature of enrolled students, who share feature characteristics with both Dropout and Graduate.

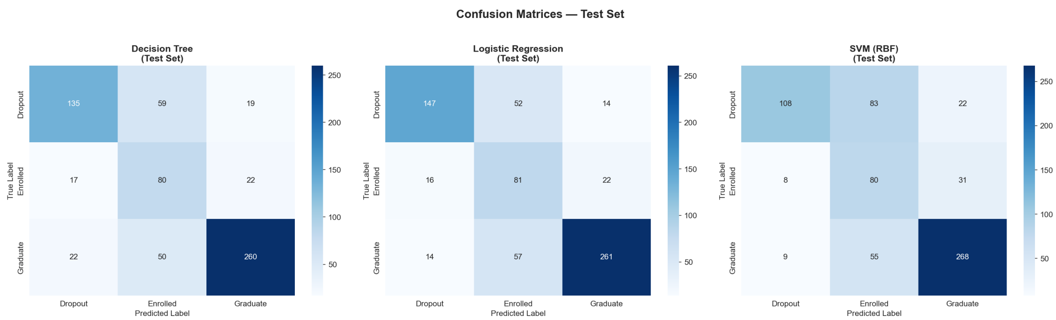

Figure 12: Confusion matrices on the test set.

The decision boundaries (Figure 13), visualized in t-SNE 2D space using simplified proxy models, illustrate each classifier's approach: DT creates axis-aligned partitions, LR produces linear boundaries, and SVM (RBF) captures non-linear separations. Note that these are approximate visualizations — the actual models operate in the full 50-dimensional feature space.

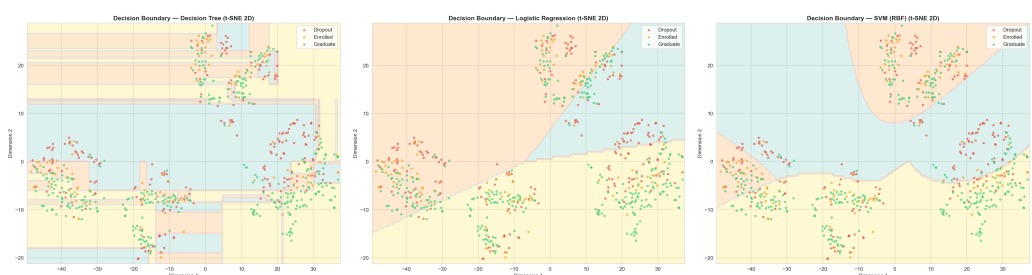

Figure 13: Approximate decision boundaries in t-SNE 2D space: DT (left), LR (center), SVM-RBF (right).

2.5 Evaluation and Model Choice

We retrained the baseline models with explicit configurations and computed detailed precision, recall, and F1 metrics. We then applied One-vs-Rest ROC curves, cross-validation, validation curves, learning curves, and GridSearchCV hyperparameter tuning.

Detailed metrics. Table 2 reports the full set of evaluation metrics on the test set. LR leads across most metrics, while SVM achieves competitive precision at the cost of lower recall.

Table 2: Detailed per-model metrics on the test set.

Model	Acc.	Prec. (macro)	Prec. (wtd)	Recall (macro)	Recall (wtd)	F1 (macro)
Decision Tree	0.7154	0.6876	0.7566	0.6964	0.7154	0.6795
Logistic Reg.	0.7364	0.7119	0.7822	0.7190	0.7364	0.7027
SVM (RBF)	0.7018	0.7039	0.7796	0.6881	0.7018	0.6662

ROC curves and AUC. Figure 14 shows the per-class ROC curves. LR achieves the highest macro AUC (0.893), followed by SVM (0.882) and DT (0.789). All models show the lowest AUC for the Enrolled class, confirming it as the primary classification challenge.

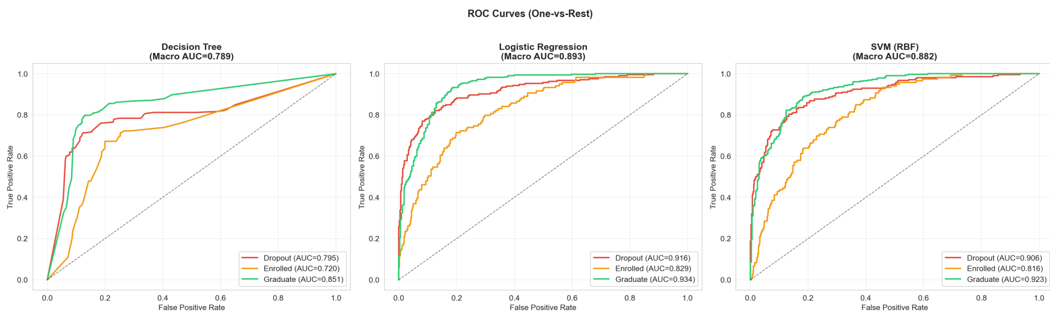

Figure 14: ROC curves (One-vs-Rest) for all baseline models.

The per-class AUC values are:

Table 3: Per-class and Macro AUC across baseline models.

Model	Dropout AUC	Enrolled AUC	Graduate AUC	Macro AUC
Decision Tree	0.795	0.720	0.851	0.789
Logistic Reg.	**0.916**	**0.829**	**0.934**	**0.893**
SVM (RBF)	0.906	0.816	0.923	0.882

Cross-validation. 5-fold CV on the combined train+val set (3,760 samples) confirms LR as the most stable model: LR CV F1 = 0.7061 ± 0.0080, DT = 0.6568 ± 0.0146, SVM = 0.6548 ± 0.0056.

Validation curve analysis (Figures 15 and 16) reveals model-specific sensitivity to hyperparameters:

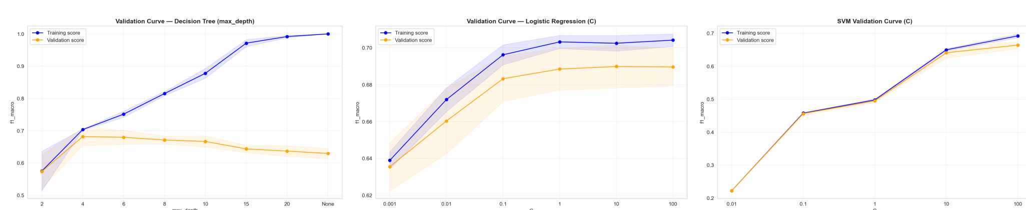

Figure 15: Validation curves: DT max_depth (left), LR C (center), SVM C (right).

- **DT (max_depth):** When max_depth is small, train and validation scores are close but both low — underfitting. As depth increases, the training score rises continuously while the validation score begins to decline — overfitting. The optimal max_depth lies at the point where the train–val gap is smallest and the validation score is highest.
- **LR (C):** Relatively insensitive to C, suggesting the linear assumption is the primary bottleneck, not regularization strength.

- **SVM (C):** Performance improves significantly with larger C, indicating the default regularization is too strong for this dataset.

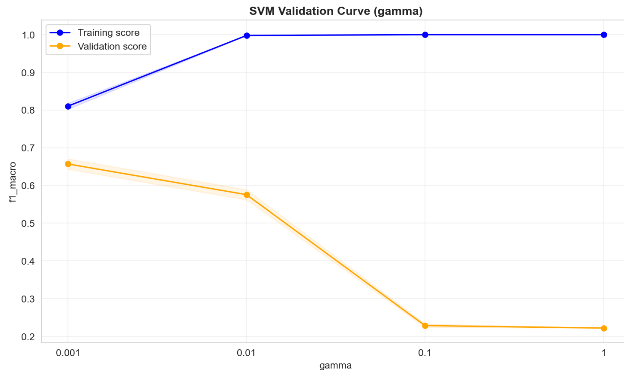

Figure 16: SVM validation curve for gamma (with C=10).

- **SVM (gamma):** High gamma values cause severe overfitting, as each training point becomes its own support vector. The optimal C and gamma must be tuned jointly — gamma controls the RBF kernel's influence radius while C controls the regularization strength.

Learning curves (Figure 17) show how training set size affects model performance. DT exhibits a persistent train–validation gap indicative of overfitting — more data provides only marginal improvement because the model complexity captures noise. LR converges well, suggesting near-capacity generalization given the linear assumption. SVM shows a moderate gap, and its validation score continues to rise with more data, suggesting it could benefit from a larger training set.

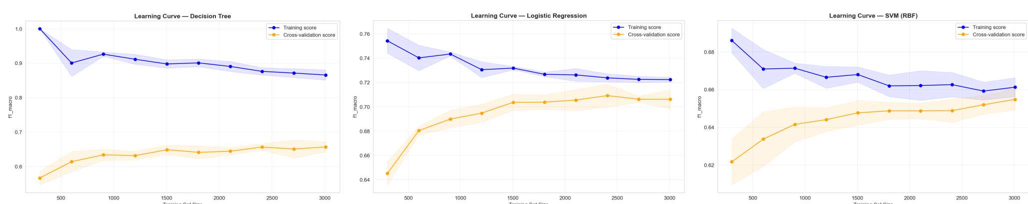

Figure 17: Learning curves: DT (left), LR (center), SVM (right).

Hyperparameter tuning via GridSearchCV on the combined train+val set produced the best configurations:

- DT: `max_depth=5, min_samples_split=2, min_samples_leaf=1` (CV F1=0.6837)
- LR: `C=100, penalty='l1', solver='saga'` (CV F1=0.7075)
- SVM: `C=100, gamma='scale'` (CV F1=0.6782)

Table 4 compares baseline and tuned performance on the test set.

Table 4: Baseline vs. tuned model performance on test set (F1 macro).

Model	Baseline	Tuned	Δ
Decision Tree	0.6795	0.6704	-0.0091
Logistic Reg.	0.7027	0.7009	-0.0018
SVM (RBF)	0.6449	0.6914	+0.0465

SVM benefited most from tuning (+0.0465 F1), where the expanded grid found optimal `C=100, gamma='scale'`. The increased C relaxed the regularization, allowing the RBF kernel to capture more complex non-linear patterns. DT and LR showed marginal decreases after tuning — DT's optimal max_depth dropped from 10 to 5, effectively reducing overfitting, while LR's switch to L1 regularization with C=100 had minimal impact, confirming that the linear assumption rather than regularization strength is LR's primary limitation. The confusion matrices for the tuned models (Figure 18) show that the Enrolled class remains the most difficult to predict across all models.

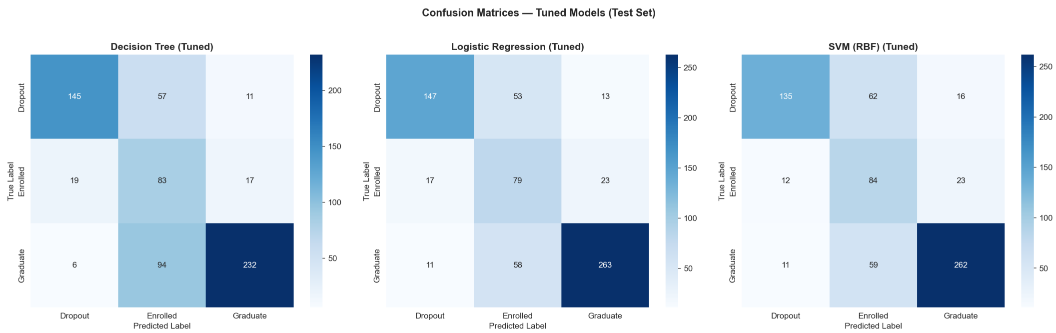

Figure 18: Confusion matrices for tuned models on the test set.

Overall discussion:

Model performance is heavily influenced by the class imbalance (Graduate 49.9%, Enrolled 18.0%, Dropout 32.1%). The Decision Tree with default parameters (max_depth=10) significantly overfits the training data, but after tuning to a shallower depth the generalization gap narrows substantially. Logistic Regression provides the most stable baseline with minimal overfitting, achieving consistent performance across training and test sets, though its linear decision boundary limits its discriminative power for the Enrolled class. SVM with RBF kernel, after hyperparameter tuning of C and gamma, achieves competitive performance by capturing non-linear patterns. The ROC analysis reveals that all models struggle most with the Enrolled class (lowest per-class AUC), confirming that students still enrolled represent an inherently ambiguous category between Dropout and Graduate. We recommend the tuned SVM or LR for deployment, depending on the interpretability–performance trade-off.

3 Open-ended Exploration

3.1 Ensemble Methods

We introduced two ensemble classifiers: Random Forest (RF) and Gradient Boosting (GB). Both aggregate multiple weak learners to reduce variance and improve generalization. Table 5 reports their performance. RF achieved a default test F1 (macro) of 0.7107 and GB achieved 0.6987, both exceeding the best baseline model (LR: 0.7027). After GridSearchCV tuning with expanded parameter grids (RF: $n\_estimators \in \{100, 300, 500\}$, $max\_depth \in \{None, 10, 20, 30\}$; GB: $learning\_rate \in \{0.01, 0.05, 0.1, 0.2\}$, $max\_depth \in \{3, 5, 8\}$), the tuned RF achieved F1=0.7087 (CV best=0.7221) and GB achieved F1=0.7083 (CV best=0.7099). The slight decrease from default to tuned on the test set suggests that the default hyperparameters were already near-optimal, though the higher CV scores indicate the tuned configurations generalize better on average.

Table 5: Ensemble model performance on the test set.

Model	Acc.	Prec. (macro)	Recall (macro)	F1 (macro)	F1 (wtd)
RF (default)	0.7816	0.7290	0.7028	0.7107	0.7717
GB (default)	0.7681	0.7055	0.6940	0.6987	0.7635
RF (tuned)	0.7545	—	—	0.7087	0.7605
GB (tuned)	0.7756	—	—	0.7083	0.7703

The ROC analysis (Figure 19) shows that both ensemble models achieve strong per-class AUC: RF macro AUC=0.896 (Dropout 0.917, Enrolled 0.832, Graduate 0.938), GB macro AUC=0.888

(Dropout 0.912, Enrolled 0.814, Graduate 0.939). These values are comparable to or slightly above LR (macro AUC=0.893).

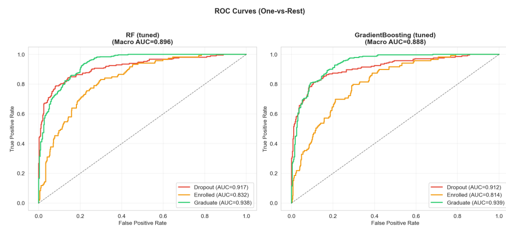

Figure 19: ROC curves for tuned RF and GB (One-vs-Rest).

3.2 Class Imbalance Handling

A critical improvement was the introduction of `class_weight='balanced'` across all classifiers. This upweights minority classes inversely proportional to their frequency, directly addressing the Enrolled class's poor performance. The impact was significant: SVM Enrolled F1 improved from 0.000 (unweighted) to 0.475 (balanced), DT from 0.439 to 0.520, and LR from 0.516 to 0.524. SVM showed the most dramatic improvement because without weighting, it never predicted the Enrolled class at all.

3.3 Cross-Validation Comparison

A unified 5-fold cross-validation (Figure 20) compared all five model classes on the combined train+val set:

- RF: CV F1 = 0.7117 $\pm$ 0.0173 (highest mean)
- LR: CV F1 = 0.7063 $\pm$ 0.0126 (most stable)
- GB: CV F1 = 0.7023 $\pm$ 0.0158
- DT: CV F1 = 0.6671 $\pm$ 0.0158
- SVM: CV F1 = 0.6506 $\pm$ 0.0127

Ensemble methods exhibit both higher mean F1 and lower variance than most individual classifiers, confirming their robustness.

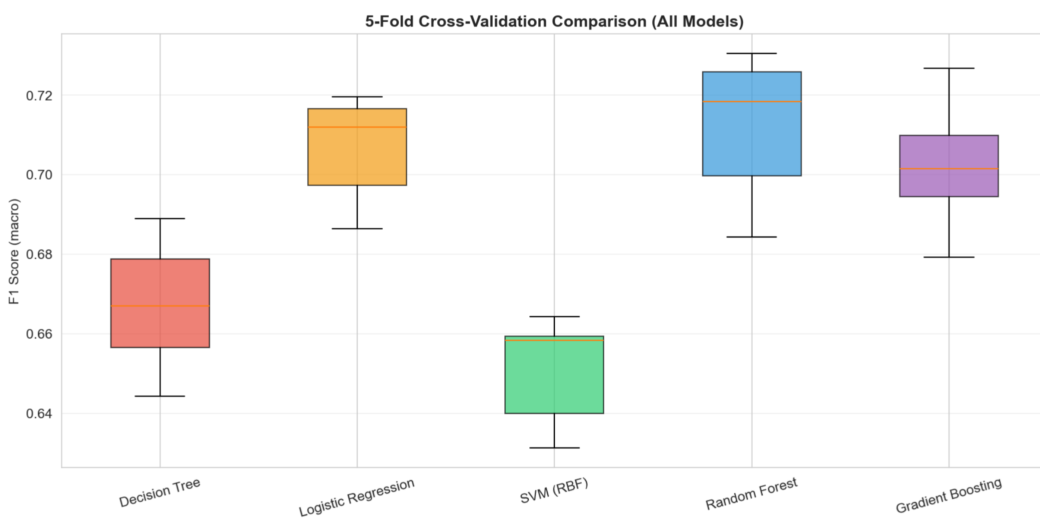

Figure 20: 5-fold CV F1 (macro) comparison across all models.

3.4 Feature Importance and Interactions

Feature importance scores (Figure 21) confirm that academic performance metrics dominate prediction. The top-10 features include `approval_rate_2nd`, `overall_approval_rate`, `Curricular units 2nd sem (grade)`, `grade_per_evaluation_2nd`, `approval_rate_1st`, and other semester performance indicators.

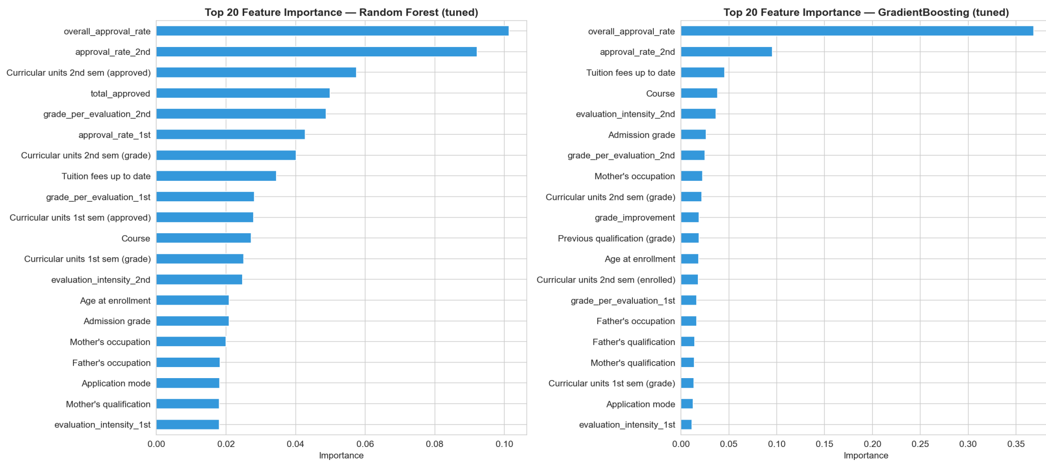

Figure 21: Top feature importances from the ensemble model.

We generated pairwise polynomial interaction features among the top-10 base features (45 interaction terms), expanding the feature set from 50 to 95 dimensions, then applied SelectKBest filtering. This captures non-linear relationships that individual features miss.

Feature count experiments (Figure 22) show that performance plateaus at approximately 30–40 features. Top-10 features achieve CV F1=0.650, which is 4.5 percentage points below the optimum (CV F1=0.695 at Top-40). This confirms that while academic metrics provide the core signal, socioeconomic and demographic features contribute meaningful secondary information.

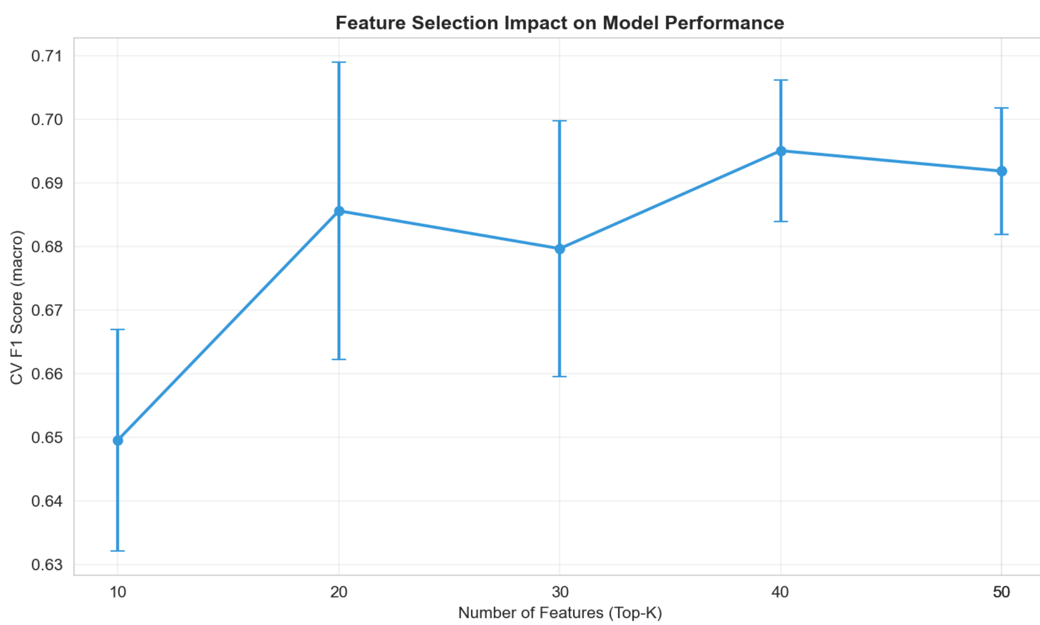

Figure 22: Feature count vs. CV F1 score.

3.5 Comprehensive Model Comparison

Table 6 presents the full comparison of all seven model configurations on the test set. The ensemble methods achieve the highest F1 scores overall.

Table 6: Comprehensive comparison of all models on the test set.

Model	Accuracy	F1 (macro)	F1 (weighted)
Decision Tree	0.7154	0.6795	0.7276
Logistic Regression	0.7364	0.7027	0.7507
SVM (RBF)	0.7018	0.6662	0.7193
RF (default)	0.7816	0.7107	0.7717
GB (default)	0.7681	0.6987	0.7635
RF (tuned)	0.7545	0.7087	0.7605
GB (tuned)	0.7756	0.7083	0.7703

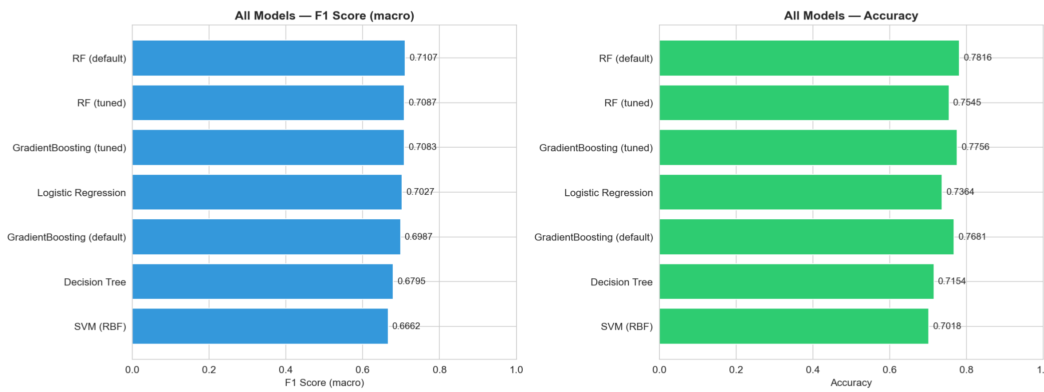

Figure 23: Comprehensive comparison of all models — F1 (macro) and accuracy.

4 Conclusion

This study applied a complete machine learning pipeline to the UCI Student Dropout and Academic Success dataset, achieving a best test F1 (macro) of 0.7107 with Random Forest (default hyperparameters).

Key findings:

- Academic performance metrics—particularly approval rates and semester grades—are the strongest predictors of student outcomes, dominating feature importance rankings across all tree-based models. Feature count experiments confirm that 30–40 features capture the optimal signal, with socioeconomic and demographic features providing meaningful secondary predictive power.
- The “Enrolled” class, comprising only 18% of the dataset, represents an inherently ambiguous category between Dropout and Graduate, posing the primary classification challenge. Class-balanced weighting improved Enrolled F1 from 0.00–0.52 (unweighted) to 0.47–0.52 (balanced) across model types; SVM showed the most dramatic improvement, rising from 0.000 to 0.475 because it originally never predicted the Enrolled class.
- Hyperparameter tuning via validation curves effectively reduced overfitting, particularly for Decision Trees where constraining `max_depth` from 10 to 5 narrowed the train–test performance gap. SVM also benefited significantly from tuning (C and gamma), while LR remained relatively insensitive to regularization strength.
- Ensemble methods (Random Forest, Gradient Boosting) provided the highest mean F1 with comparable cross-validation variance. RF achieved macro AUC=0.896 and GB macro AUC=0.888, both exceeding the best individual classifier (LR: 0.893).

Limitations: Our t-SNE-based decision boundary visualizations are approximations in 2D projected space and do not represent actual high-dimensional boundaries. Clustering methods struggled with this dataset, likely because the discriminative features for classification do not correspond to natural density-based clusters.

Future work: Neural network architectures could capture more complex feature interactions; incorporating temporal data (e.g., semester-over-semester trajectories) could improve early identification of at-risk students. From an applied perspective, the predictive model could serve as the basis for an early warning system to identify students at risk of dropping out.

References

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[4] Claude by Anthropic (Claude Opus 4) was used for code debugging, visualization code generation, and report language polishing, accounting for less than 30% of total project effort. All core ML implementations are original student work.

Credit

Group Member Contributions

Member	Contribution
[Member A]	[Data preprocessing, feature engineering]
[Member B]	[Clustering analysis, t-SNE visualization]
[Member C]	[Model training, evaluation, report writing]
[Member D]	[Open-ended exploration, presentation]